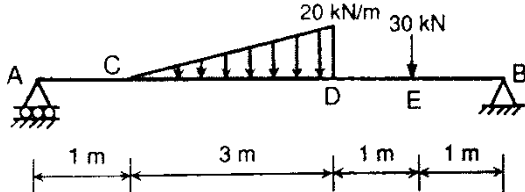
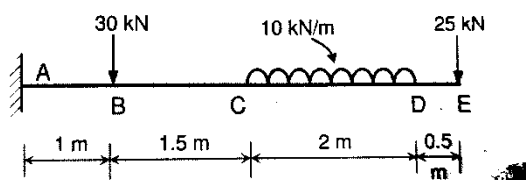


Instructions to the Students:

1. All the questions are compulsory.
2. Figures to right indicates full marks.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain Stress Strain Graph for Mild Steel	CO-I	06
B) Derive the equation of Volumetric Strain for Tri axial loading	CO-I	06
C) A steel rod of 20mm diameter and 300mm long is enclosed centrally inside a copper tube of external diameter 30mm and internal diameter 25mm. the ends of tube and rod are brazed (fixed) together. And composite bar is subjected to an axial pull of 40kN. Find the stresses developed in rod and tube . Also find the extension of rod. Take $E_s = 200 \frac{GN}{m^2}$ $E_c = 100 \frac{GN}{m^2}$	CO-I	06
Q.2 Solve Any one of the following.		12
A) Draw shear force and Bending moment Diagram for the beam as shown in fig.1, also locate the position and Magnitude of the maximum Bending moment.	CO-II	12
 Fig.1		
B) Draw shear force and Bending moment Diagram for the cantilever beam as shown in fig. 2.	CO-II	12
 Fig.2		
Q. 3 Solve Any Two of the following.		12
A) Derive Flexural formula	CO-III	06
B) A cantilever beam of uniform rectangular cross section 150mm x 300mm uniformly distributed load of 20kN/m over its span of 2m. Determine the bending stress anywhere in the beam.	CO-III	06
C) Find the power that can be transmitted by shaft of 60mm diameter and rotating at 180 r.p.m., If the permissible stress is $85 \frac{N}{mm^2}$.	CO-III	06

Q.4 Solve Any Two of the following.		12
A) Define Strut, column and slenderness ratio	CO-III	06
B) Calculate the safe compressive load on a hollow C.I. column (one end fixed and other hinged) of 150mm external diameter, 100mm internal diameter and 10m length. Using Euler's formula with factor of safety of 5 and $E=95\text{GPa}$	CO-III	06
C) The external and internal diameter of a hollow C.I. column are 5cm and 4cm respectively. If the length of its column is 3m and both of its ends are fixed. Determine the crippling load using Rankine's Formula. Take $\sigma_c=550\frac{N}{mm^2}$ A= 1/1600 in Rankine's Formula.	CO-III	06
Q. 5 Solve Any Two of the following.		12
A) The Tensile Stresses at a point across two mutually perpendicular planes are $100\frac{N}{mm^2}$ and $40\frac{N}{mm^2}$. Find the graphically the normal and tangential stresses on a oblique plane inclined at 30° with the major principal planes. Also find the resultant stress and angle of obliquity.	CO-IV	06
B) A Bar of uniform cross section 60mm x 80mm is subjected to an axial tensile force of 600kN applied at each end of the bar determine the normal and shear stress acting on a plane inclined at 35° to the line of action of force.	CO-IV	06
C) A point in a strained material is subjected to a tensile stress of $120\frac{N}{mm^2}$ and compressive of $80\frac{N}{mm^2}$ acting at right angles to each other. Find the normal Stress, tangential stress and the resultant on a plane inclined at angle of 30° with the compressive stress.	CO-IV	06

***** End *****